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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 1: Introduction to Transportation Systems-I

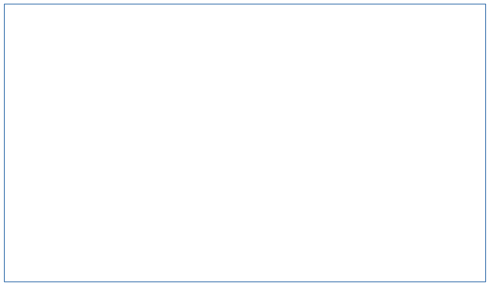
Prof. Bhola Ram Gurjar

Dept. of Civil Engineering



Contents

- Evolution of Transportation modes
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  - Animals as Transport modes
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  - The Gasoline Bus
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  - Aircrafts
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Evolution of Transportation Systems



First form of Transport on land was, of course, WALKING !!

Evolution of Transportation Systems

Animals as Transport modes

Animals such as Donkeys, Mules and Horses were used as transport modes for person and goods







Using Animals for Transport

Evolution of Transportation Systems

Primitive Log Rafts, 8000 BCE





A tree trunk floating downstream with the current, might be the first stimulus to man's inventiveness in this direction



The Oldest Boat found (Pesse Dugout Canoe ≈8000 BCE) Source: (http://www-labs.iro.umontreal.ca/~vaucher/History/Ships/Prehistoric_Craft/index.html)

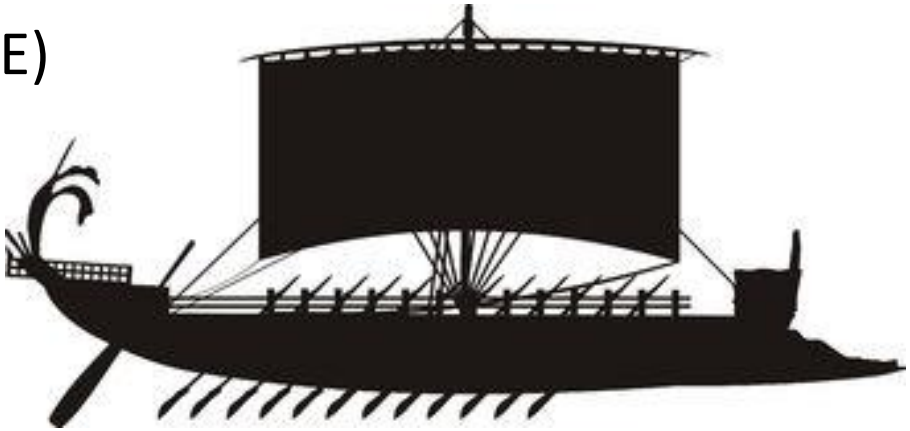


Primitive Log Raft Source: (http://www-labs.iro.umontreal.ca/~vaucher/History/Ships/Prehistoric_Craft/index.html)

Source: (James Hornell (1946), Water Transport- Origins and Early Evolution)

Evolution of Transportation Systems

The Egyptian Era (6000 BCE - 3000 BCE)



- Egyptians built some of the earliest boats ever recorded.
- The first ones were made from papyrus reeds and propelled by rowing.
- Mainly used for trade.

By 2500 BCE, wooden boats that could cross Oceans were built.

Source: (James Hornell (1946), Water Transport- Origins and Early Evolution)

Evolution of Transportation Systems

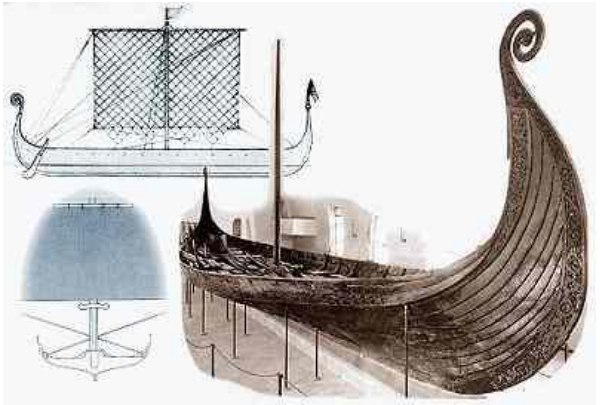
The Viking Ships

Two different classes of Viking era ships

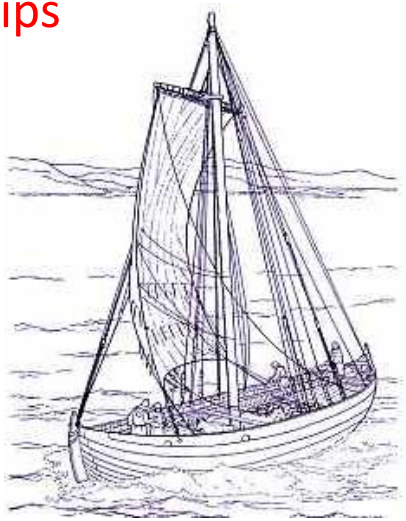
War ships and Merchant ships

- One of the greatest technical and artistic achievement.

Fast and Narrow war ships had the strength to survive ocean crossings as well as allowing navigation in very shallow water (having a draft of as little as 50 cm (20 inches).



Viking warships called "langskip"

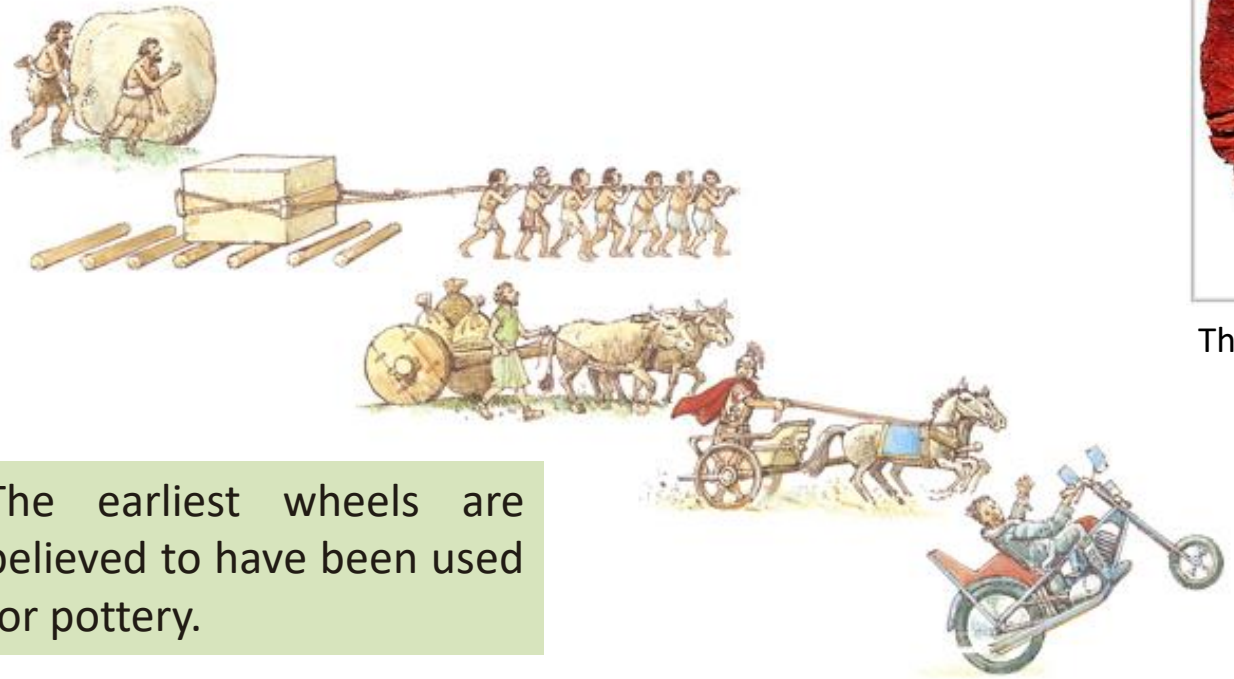


Viking merchant ships called "knorr"

Source: (http://www.hurstwic.org/history/articles/manufacturing/text/norse_ships.htm)

Evolution of Transportation Systems

Invention of Wheel, 3500 BC





The Wheel of Mesopotamia, 3500 BC

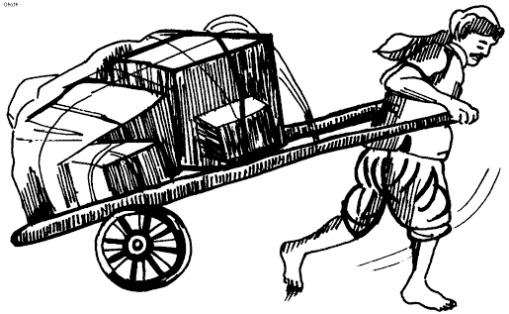
The earliest wheels are believed to have been used for pottery.

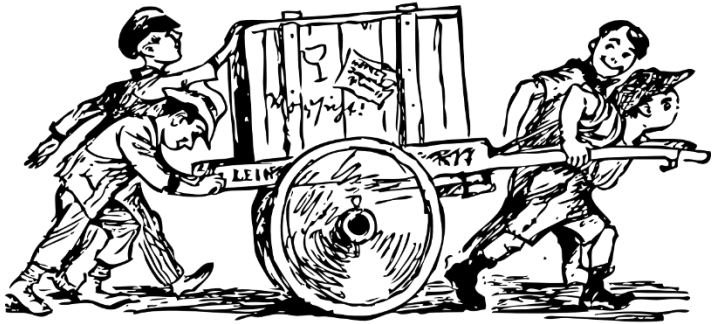
Evolution of Transportation Systems

Wheeled Carts

Wheeled carts were pulled for transport of goods.





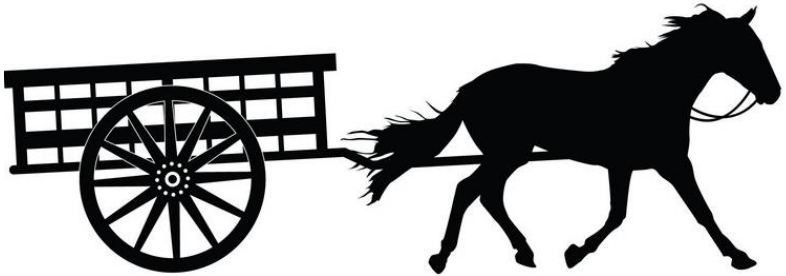


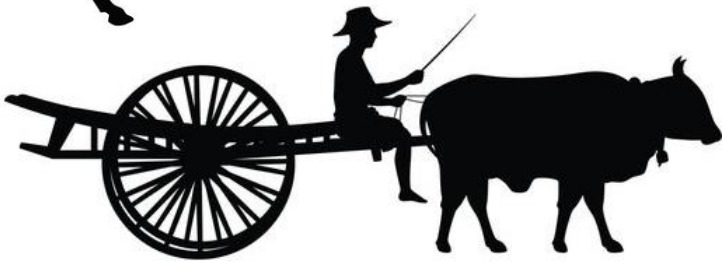
Evolution of Transportation Systems

Animal Carts

Animals were trained to pull carts



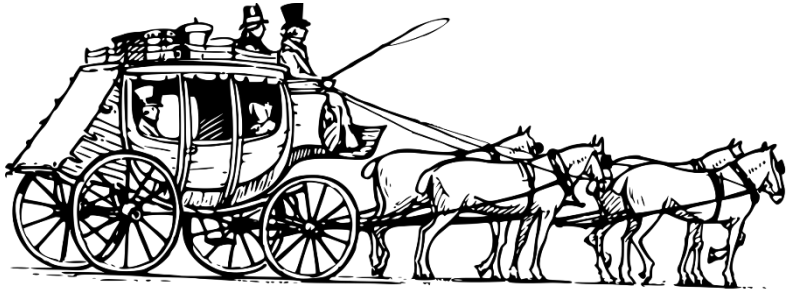




Evolution of Transportation Systems

Wagons, Coaches and Carriages

Horse-driven wagons, coaches and carriages as superior modes of transport

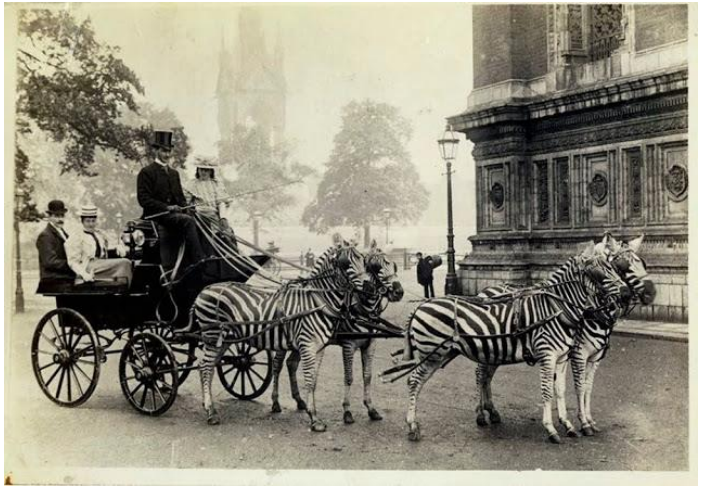






Evolution of Transportation Systems

Wagons, Coaches and Carriages





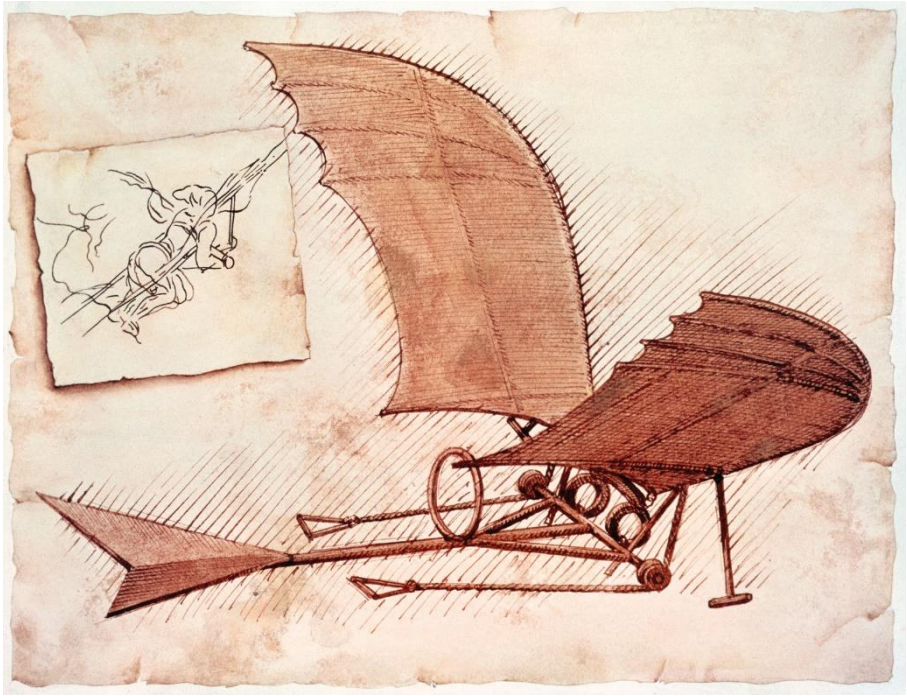
Humans even tried to domesticate Zebras

Zebra carts been used by humans in the 1800s

Source: (http://www.urban75.org/brixton/history/zebra-taxi-cab.html)

Evolution of Transportation Systems

The Dream of Flying “Leonardo da Vinci’s Flying Machine”, 1490



In about 1490 Leonardo da Vinci drew plans for a flying machine

Source: (https://www.britannica.com/technology/history-of-flight)

Evolution of Transportation Systems

The First True Automobile, 1769

In 1769 Nicolas-Joseph Cugnot built a three-wheeled steam-driven vehicle.



1769 Cugnot

Source: (Encyclopædia Britannica, Inc.)

- Because of the heavy weight of the steam chamber in the front, it had a tendency to tip over when not hauling cannons, which was what it was designed to do.
- Steam buses were running in Paris about 1800.

Source:(Encyclopædia Britannica, Inc. https://www.britannica.com/technology/automobile/History-of-the-automobile)

Evolution of Transportation Systems

Ballooning, 1783



Source: (Encyclopædia Britannica, Inc.)

French aeronauts Jacques-Alexandre-Cesar Charles and Marie-Noel Robert made the first manned ascent in a gas balloon, Dec. 1, 1783

Source: (https://www.britannica.com/technology/balloon-flight/Historical-development)

Evolution of Transportation Systems

The "Age of Canals" (18th Century)

The first powerful motor of the industrial age

- Facilitated flows of goods, unprecedented exchanges between regions, specialization of labor, and access to more distant energy and raw material resources.

- Local fuelwood shortages were resolved by substituting with a new higher energy density fuel, "coal" and was made possible by canals.



The Manayunk Canal section of Philadelphia started in 1800s and were operational upto the 20th century. (Source: https://www.ushistory.org/us/25a.asp)

Source: (Arnulf Grubber and Nebojsa Nakicenovic (1991))

Evolution of Transportation Systems

Draisienne, 1817, The First Two-wheeler

Draisienne, also known as Dandyhorse was invented by Karl Drais in 1817.



- First two wheeled rider-propelled machine.
- The machine was made of wood, and the seated rider propelled himself by paddling his feet against the ground.

Source:(Encyclopædia Britannica, Inc. https://www.britannica.com/technology/bicycle#ref183542)

Evolution of Transportation Systems

Steam as Fuel, (1819)



- Steam-hauled public railways began with the Stockton and Darlington Railway in 1825.
- Reduced time used to transport goods and allowed for increased specialization.

SS Savannah, the Hybrid ship using Steam engine and paddles using Wind power

The SS Savannah became the first steamship to cross the Atlantic Ocean

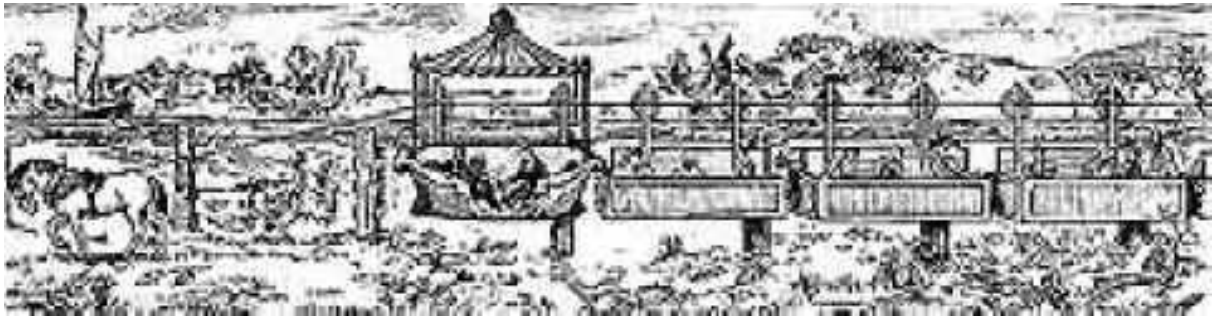
(Source: https://transportationhistory.org/2017/06/20/today-in-transportation-history-1819-the-first-steamship-to-cross-the-atlantic/)

Evolution of Transportation Systems

The Cheshunt Railway, 1825, World's First Monorail

The world's first Monorail was drawn by a horse.

- First passenger drawn Monorails in the world.
- The design was an overhead track from which carriages were suspended, drawn by a single horse.



Source:(The Monorail Society, https://www.monorails.org/tMspages/History.html)

Evolution of Transportation Systems

The Omnibus, 1826



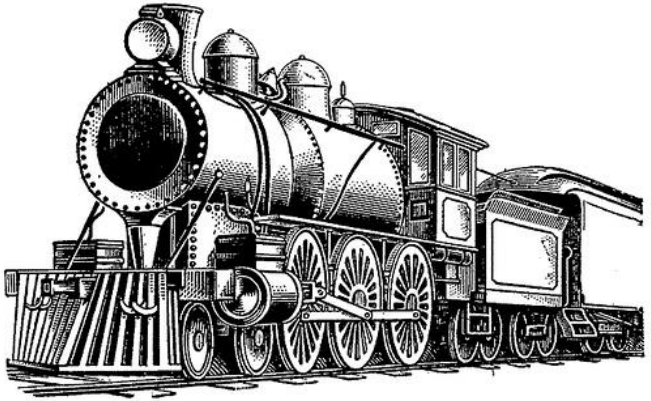
The Omnibus in 1826 was the first land based innovation in public transportation system

- These were horse drawn passenger wagons

Source:(https://gogocharters.com/blog/history-of-public-bus-transportation/#:~:text=The%20first%20person%20to%20propose,routes%20in%20Paris%20in%201662.)

Evolution of Transportation Systems

The “Great Railway Era” (1830s -1930s)

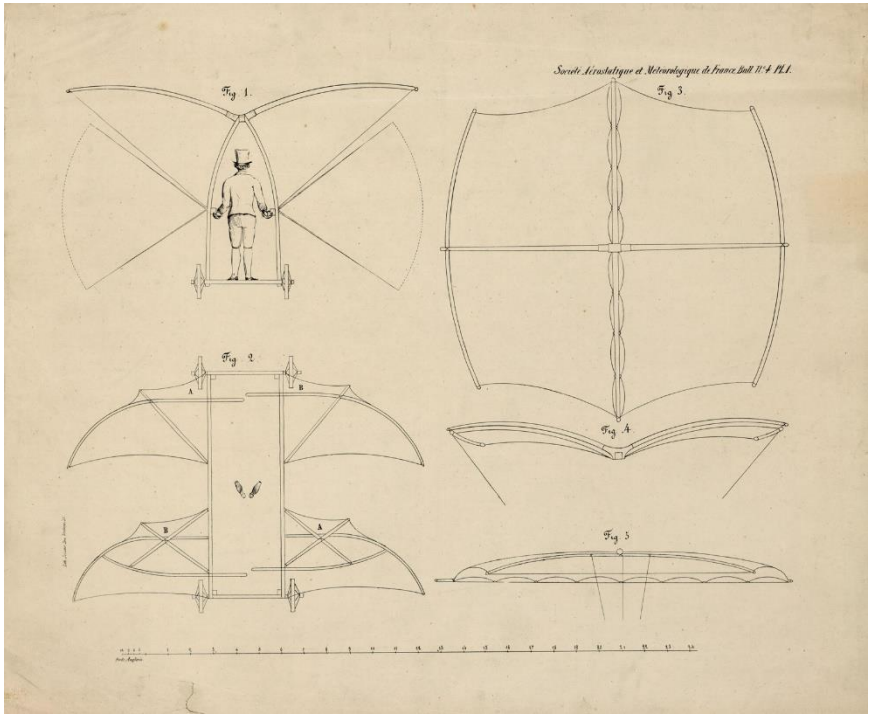


- The first railways were constructed in the 1830s.
- Extended the range, speed, and productivity levels of transport, earlier achieved by the canals.
- With railways, a new era of coal, steam, steel, and the telegraph began.
- The Great Railway era lasted till the 1930s.

Source: (Arnulf Grubber and Nebojsa Nakicenovic (1991))

Evolution of Transportation Systems

The “First Successful Manned Flight, 1853



English aeronautic pioneer George Cayley designed a glider (shown in the drawing) in 1853, recorded as the first successful manned flight.

Source: (Library of Congress, Washington, D.C. (neg. no. LC-DIG-ppmsca-02521))

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight)

Evolution of Transportation Systems

The Velocipede, 1860s



The Velocipede, colour lithograph by Nathaniel Currier and James M. Ives, 1869

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/velocipede#/media/1/624894/16899)

Velocipede, version of the bicycle reinvented in the 1860s by the Michaux family of Paris.

- The bicycle version had iron and wood construction and the lack of springs earned it the nickname “boneshaker”.
- It was driven by pedalling cranks on the front axle.

Evolution of Transportation Systems

The Philadelphia Centennial, First Steam driven Monorail, 1876



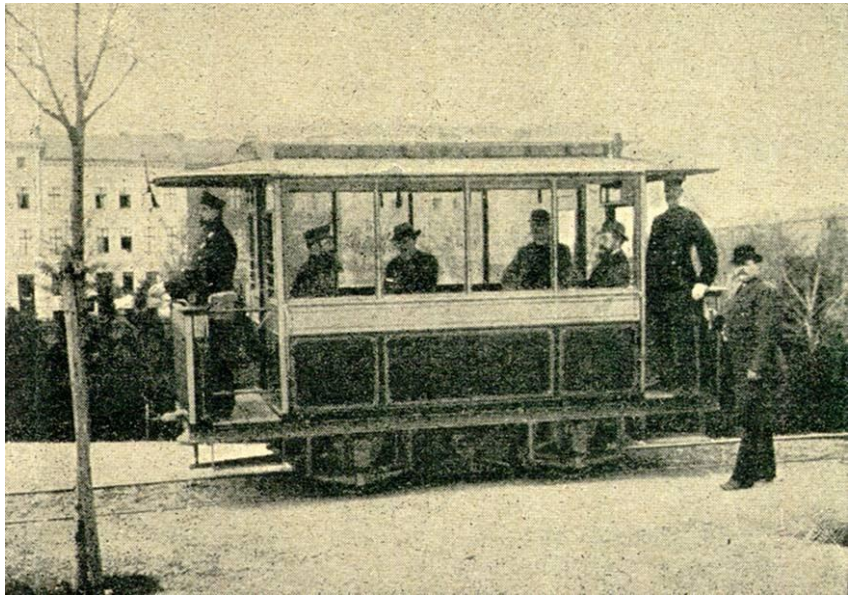
The first steam driven Monorail, demonstrated at the United States Centennial Exposition, 1876.

- The ornately designed double-decker vehicle had two main wheels, the rear one driven by a rotary steam engine.

Source:(The Monorail Society, https://www.monorails.org/tMspages/History.html)

Evolution of Transportation Systems

The World's First Electric Tram, 1876



The world's first electric tram, The Grob-Lichterfelde Tramway

Source:(https://engre.co/about/achievements/the-first-electric-tram-in-the-world/)

The World's first electric tram was exhibited in the 1879 Berlin Commercial Exposition.

- The tram had a driver seat, 3 little carriages, each for 3 people capacity.
- Ran in Direct current (approx. 149 volts).

Evolution of Transportation Systems

Penny-farthing Bicycle, 1883



Type of bicycle that was most used between the time of the introduction of the French mass produced chain-driven bicycle Boneshaker in 1860s and the English “safety bicycle” in 1880s

James Starley's “penny-farthing” bicycle, 1883
Source: (https://www.britannica.com/technology/penny-farthing#/media/1/450204/4516)

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/bicycle/From-boneshakers-to-bicycles#ref183554)

Evolution of Transportation Systems

The Safety Bicycle, Late 1885



The Rover Safety bicycle developed by John Kemp Starley during the 1880s is the forerunner of all modern bicycles

- Had all the basic features of standard modern bicycles, including chain drive, which meant that both wheels could be the same size.

The Rover Safety Bicycle, 1880s Source: (https://www.bbc.co.uk/ahistoryoftheworld/objects/u76Sy05eSNi0zXeC5vDPmg#:~:text=all%20modern%20bicycles.-,The%20Rover%20Safety%20Bicycle%20was%20developed%20by%20J.K.,forerunner%20of%20all%20modern%20bicycles.&text=This%20form%20of%20bicycle%20revolutionised,important%20form%20of%20personal%20transport.)

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/bicycle/From-boneshakers-to-bicycles#ref183554)

Evolution of Transportation Systems

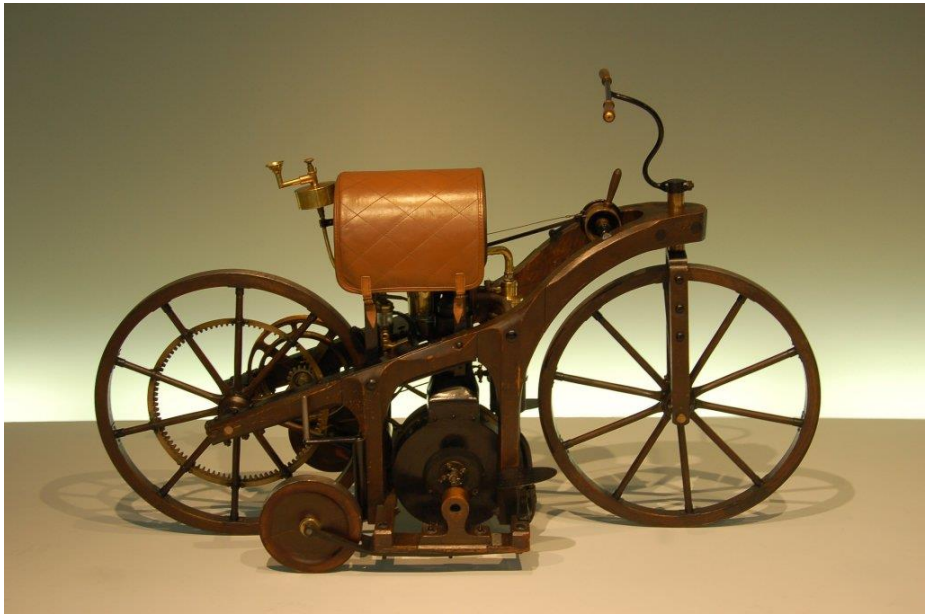
Coal as Fuel, 1880s

- During the late 1800s to the 1950s, coal was the world's primary heating and transportation fuel.
- In 1880, coal powered a steam engine attached to the world's first electric generator.
- By the late 1800s, a new form of fuel was catching on: Petroleum



Evolution of Transportation Systems

“Daimler Reitwagen” The First True Motorcycle (1885)



Daimler’s and Maybach’s first motorcycle. The ‘Reitwagen’

Source: (https://mercedesbenzstories.wordpress.com/2015/01/09/the-story-behind-mercedes/)

- Two German inventors -Gottlieb Daimler and Wilhelm Maybach managed to produce first modern model of motorcycle.
- Attached small petroleum/gasoline based combustion engine to the wooden bicycle frame.

Source: (http://www.bicyclehistory.net/motorcycle-history/first-motorcycle/)

Evolution of Transportation Systems

The First Gasoline Car, 1885



The Original “Benz Patent Motor Car”, 1886

Source: (https://www.daimler.com/company/tradition/company-history/1885-1886.html#:~:text=1885%E2%80%931886.-,The%20first%20automobile,on%20New%20Year's%20Eve%201879)

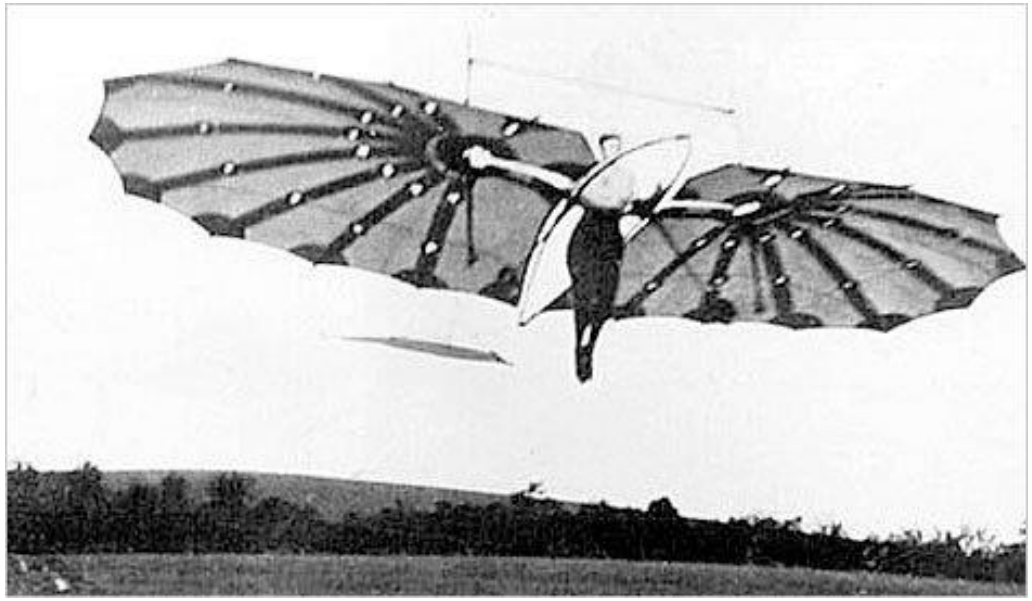
Benz ran his first car in 1885, Daimler in 1886, Hammel in 1888.

- The first stationary gasoline engine developed by Carl Benz was a one-cylinder two-stroke unit.

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/automobile/Early-electric-automobiles)

Evolution of Transportation Systems

Pilcher Hawk “The Monoplane glider”, 1896



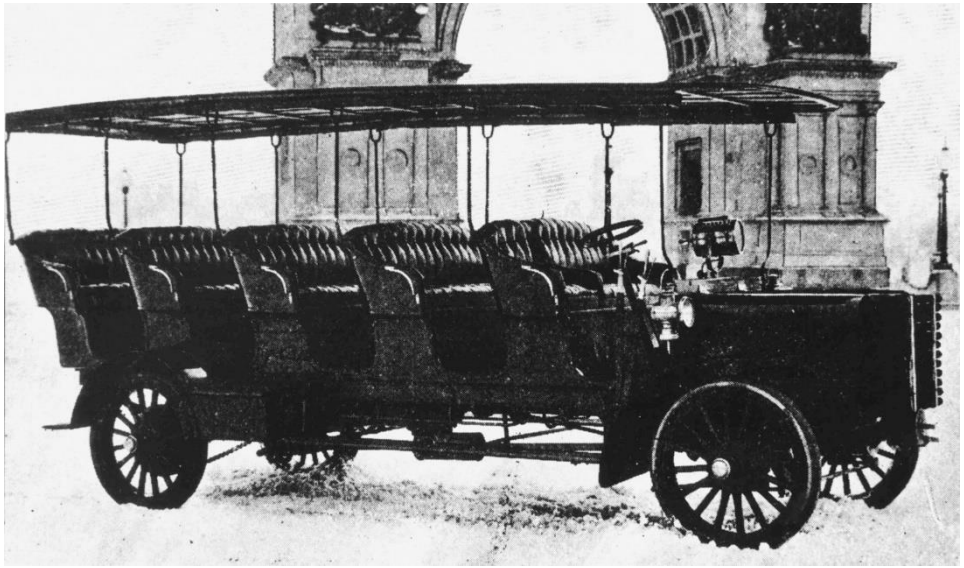
Source: (Encyclopædia Britannica, Inc.)

In 1896, Percy Sinclair Pilcher designed, built, and flew “the Pilcher Hawk”, a monoplane glider with birdlike wings.

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight)

Evolution of Transportation Systems

The Gasoline powered Bus (1900s)



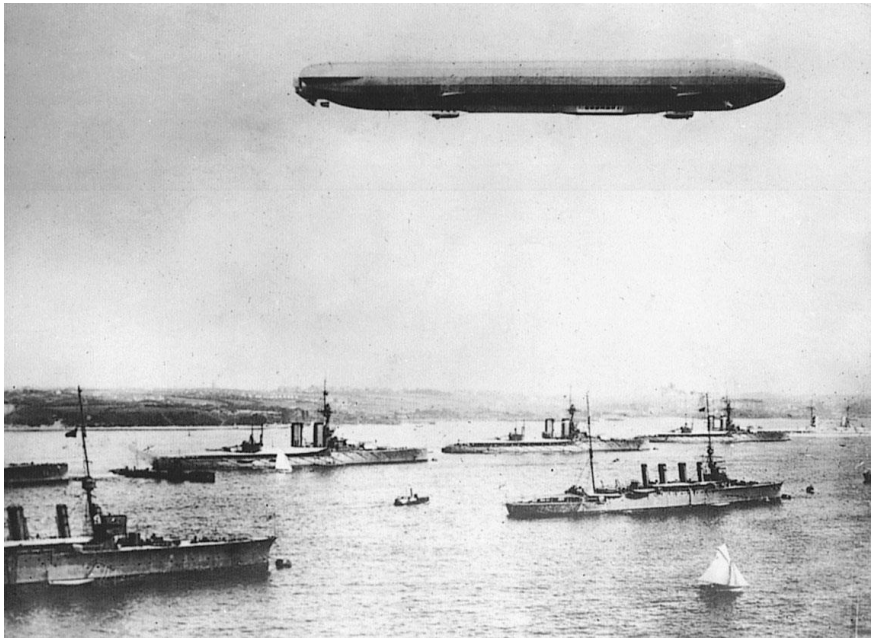
These buses had a nominal seating capacity of 20 with a four-cylinder gasoline engine developing 40 horsepower at street speeds of up to 32 km (20 miles) per hour.

A 20-passenger 40-horsepower bus built by Mack Trucks for sightseeing in Brooklyn's Prospect Park, 1900.

Source: (The Mack Trucks Historical Museum, https://www.britannica.com/technology/bus-vehicle#/media/1/86028/162136)

Evolution of Transportation Systems

Airships or Blimps, 1900s



A zeppelin flying over the harbour at Kiel, Ger. during World War I.

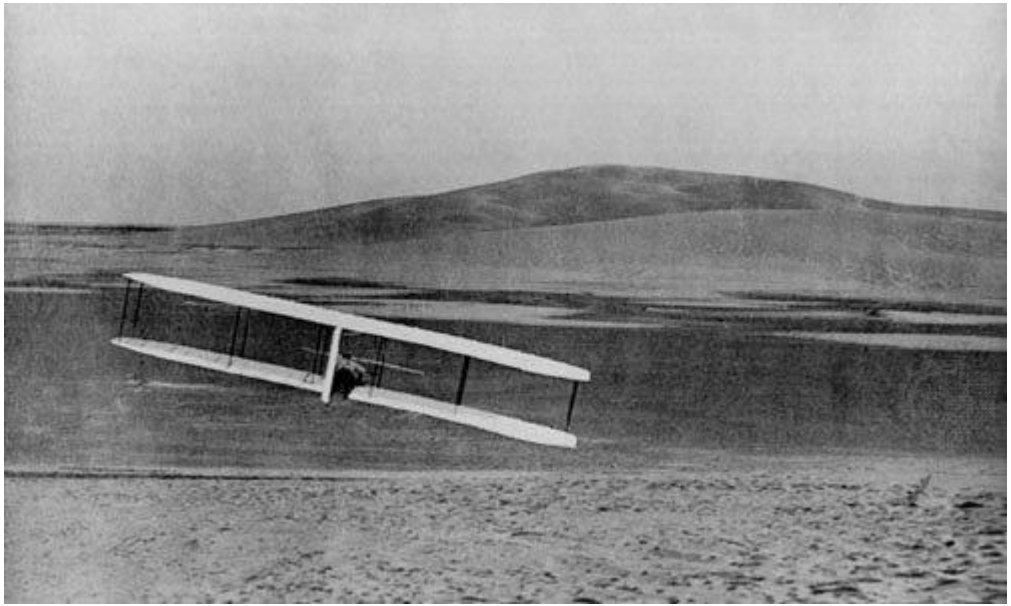
Source: (Encyclopædia Britannica, Inc.)

Source: (https://www.britannica.com/biography/Ferdinand-Graf-von-Zeppelin)

- Began to be built in early 1900s.
- Inflated by hydrogen gas to keep them aloft.
- Had engines with propellers as well as flaps to control the direction and speed of flight.

Evolution of Transportation Systems

Wright Glider, 1902



First Fully controllable Glider by Wright brothers in 1902

Wright brothers' first fully controllable glider, North Carolina, October 24, 1902.

Source: (Wright State University, Archives & Special Collections)

Source: (https://www.britannica.com/technology/history-of-flight)

Evolution of Transportation Systems

Wright Flyer, 1905



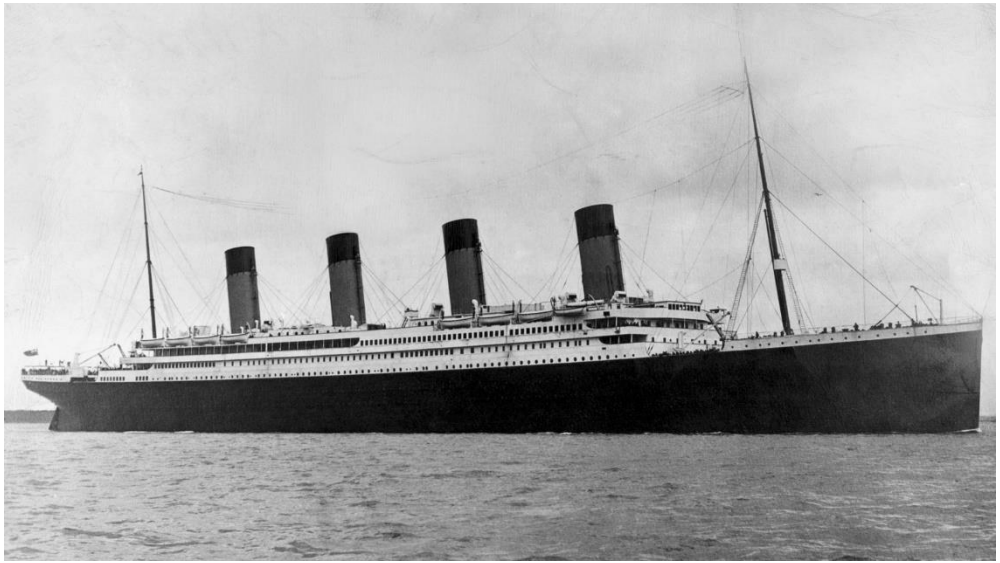
First practical Flying machine by the Wright brothers, “The Wright Flyer” in 1905

The Wright brothers' first practical flying machine, with Orville Wright at the controls, passing over Huffman Prairie, near Dayton, Ohio, October 4, 1905. Source: (Library of Congress, Washington, D.C. (digital file no. 00658u))

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight)

Evolution of Transportation Systems

The Unsinkable Titanic, 1912



The RMS Titanic

Source: (https://www.britannica.com/topic/Titanic#/media/1/597128/7788)

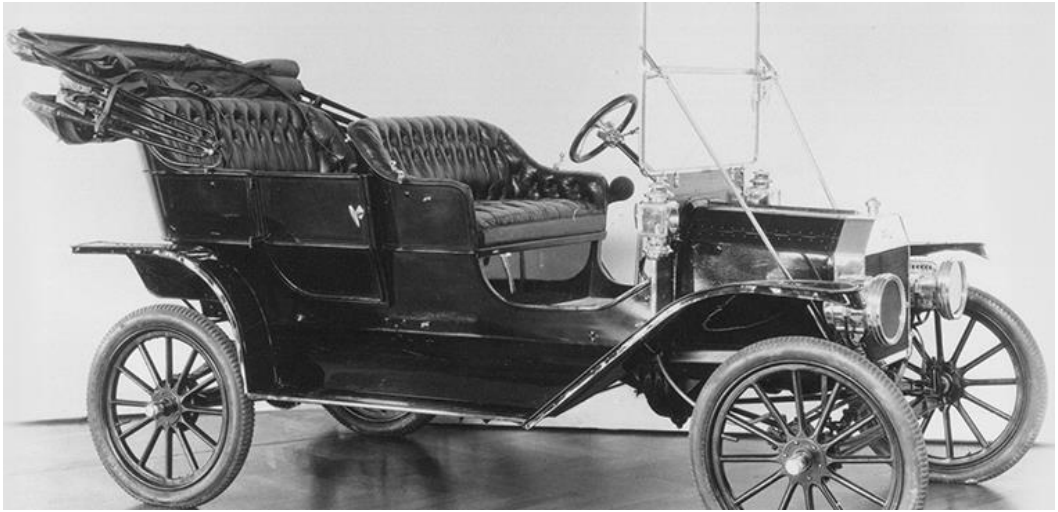
On April 10, 1912, The RMS Titanic embarked on its maiden voyage from Southampton to New York City

- The RMS Titanic was World's largest passenger ship at that time and was thought of as an Unsinkable ship.

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/story/the-unsinkable-titanic)

Evolution of Transportation Systems

Automobile from a Luxury to a Necessity



Model T, The Universal car

(Source: https://corporate.ford.com/articles/history/the-model-t.html)

Introduced to the world in 1908 by Henry Ford, Model T was one of the first mass production vehicles.

- Turned the automobile from a luxury asset into a necessity by making it cheap, versatile, and easy to maintain.

Source: (Encyclopædia Britannica Inc. https://www.britannica.com/technology/Model-T)

Evolution of Transportation Systems

Vickers Vimy, 1919



Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight)

The Vickers Vimy plane used by John Alcock and Arthur Brown in the first nonstop transatlantic flight in 1919.

Evolution of Transportation Systems

Douglas DC-3 Passenger aircraft, 1935



Douglas DC-3 passenger aircraft, which first flew in 1935.

The DC-3 dominated the airline business until the end of World War II.

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight)

Evolution of Transportation Systems

The Hovercraft, 1959



The Mountbatten class hovercraft

Source: (https://www.britannica.com/technology/Hovercraft#/media/1/1558876/116738)

The Hovercraft, are a series of British-built and British-operated air-cushion vehicles (ACVs) that for 40 years (1959–2000) ferried passengers and automobiles across the English Channel between southern England and northern France

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/Hovercraft)

Evolution of Transportation Systems

Bell Helicopter 206-B (Jet Ranger), 1966



The Bell Helicopter 206-B (Jet Ranger) was introduced in 1966.

- The turboshaft helicopter could carry four passengers and was often used for deployment of firefighters

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight#/media/1/210191/51211)

Evolution of Transportation Systems

Concorde supersonic aircraft, 1969



Concorde, the first supersonic passenger-carrying commercial airplane (or supersonic transport, SST), built jointly by aircraft manufacturers in Great Britain and France.

- Concorde first flew in 1969 and entered commercial service in 1976 and operated till 2003.

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight#/media/1/210191/51215)

Evolution of Transportation Systems

Tupolev Tu-144 (Breaking the sound barrier), 1969



On June 5, 1969, Tu-144 became the first passenger aircraft to break the sound barrier.

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight#/media/1/210191/50964)

Evolution of Transportation Systems

Airbus 320, 1988

Airbus A320 short- to medium-range jetliner, first flew in 1987 and went into commercial service in 1988.



- The aircraft typically accommodates 150 passengers.
- Its success led to a family of derivative aircraft of varying passenger capacities, including the A318, A319, and A321.

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight#/media/1/210191/51439)

Evolution of Transportation Systems

Bombardier Global Express, 1999



Bombardier Global Express long-range business jet with twin engines.

- Bombardier had range of more than 11,100 km (6,900 miles) and can approach the speed of sound.

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/history-of-flight#/media/1/210191/51232)

Evolution of Transportation Systems

Maglev, 2003



- A maglev vehicle rides on an air cushion created by electromagnetic reaction between an on-board device and another embedded in its guideway.
- Propulsion and braking are achieved by varying the frequency and voltage of a linear motor system embodied in the guideway and reacting with magnets on the vehicles.

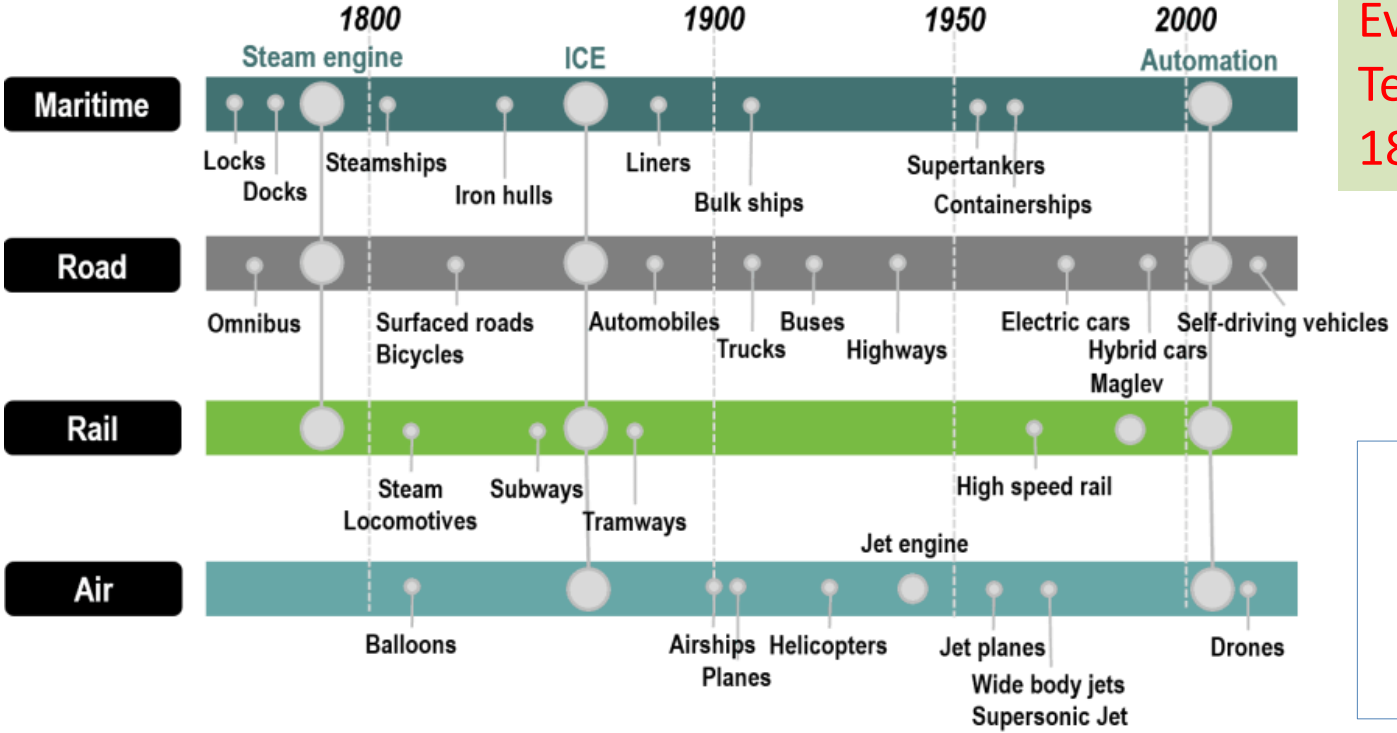
Trial run of the Maglev Transrapid 06, Germany

Source: (https://www.britannica.com/technology/railroad/South-Korea-Taiwan-and-China#/media/1/489715/3787)

Source: (Encyclopædia Britannica, Inc. https://www.britannica.com/technology/railroad/South-Korea-Taiwan-and-China#ref919226)

Evolution of Transportation Systems

Timeline



Evolution of Transport Technology since the 18th Century

Source: (Jean-Paul Rodrigue (2020), Transport Geography)

Conclusion

- The Transportation systems have changed drastically after the Industrial revolution.

- The use of Coal and petroleum as major sources of fuels, had a great impact on the environment and the natural resources.

- With the technological developments and use of natural resources as fuel, there is a need to reduce the impacts of emissions on the environment.

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Thank You